Matthew J. Payne

Director of the Minor Planet Center

Smithsonian Astrophysical Observatory, Center for Astrophysics | Harvard & Smithsonian

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Profile

I am an astrophysicist with research interests in the observations and dynamics of exoplanets and Solar System bodies. I am interested in the development of algorithms and open-source software for the efficient processing of large astronomical data sets, and I am enthused by the cross-disciplinary work of the planetary defense community.

Dynamics · Formation and evolution of small-body populations · Planetary defense · Orbit fitting · Software development for astrophysics · Data processing and distribution

Appointments

2022–present **Director, Minor Planet Center**
Smithsonian Astrophysical Observatory, Cambridge, MA
I lead a team of 12 astronomers and software developers at the Minor Planet Center (MPC), the single worldwide location for the receipt and distribution of positional measurements of minor planets, comets and outer irregular natural satellites of the major planets, and for the identification, designation and orbit computation of all of these objects.
I lead the design and delivery of advanced tools for processing and cataloguing near-Earth objects, asteroids, comets and the natural satellites of the major planets, and manage the collection and analysis of data from a worldwide network of observatories.
I write grant proposals to fund both new research and the expansion of the MPC's activities, and I carry out research on the observations and dynamics of Solar System bodies.

2021–2022 **Acting Director, Minor Planet Center**
Smithsonian Astrophysical Observatory, Cambridge, MA

2017–2021 **Project Scientist, Minor Planet Center**
Smithsonian Astrophysical Observatory, Cambridge, MA
Led the MPC's efforts to design and deliver new software tools for processing and cataloguing astrometric observations of Solar System objects.

2014–2017 **Research Scientist**
Smithsonian Astrophysical Observatory, Cambridge, MA
Research on the observation and dynamics of exoplanets and Solar System objects.

2012–2014 **Postdoctoral Researcher (theory)**
Harvard-Smithsonian Center for Astrophysics, Cambridge, MA
Research on the observation and dynamics of exoplanets and Solar System objects.

2009–2012 **Postdoctoral Researcher (theory)**
University of Florida, Gainesville, FL
Research on the observation and dynamics of exoplanets.

2005 **Predoctoral Researcher**
Institute of Astronomy, University of Cambridge, UK
Globular cluster analysis.

2001–2004 **Management Consultant**
Mars & Co, London, UK
Quantitative analysis for a large telecommunications client.

Education

2005–2009 **PhD in Astronomy**
Institute of Astronomy, University of Cambridge, UK
Thesis: On the Formation and Evolution of Planetary Systems

2004–2005 **Certificate of Advanced Study in Mathematics (Part III Mathematics)**
DAMTP, University of Cambridge, UK
Theoretical physics and astrophysics

1997–2001 **BA and MPhys in Physics**
University of Oxford, UK

Publications

65 refereed papers, 9 as first author (NASA ADS). Full list: matthewjohnpayne.github.io/publications.

Recent first-author papers

1. Payne, M. J., Veras, D., Gänsicke, B. T., Holman, M. J. (2017). The fate of exomoons in white dwarf planetary systems. *Monthly Notices of the Royal Astronomical Society* 464, 2557.
2. Payne, M. J., Veras, D., Holman, M. J., Gänsicke, B. T. (2016). Liberating exomoons in white dwarf planetary systems. *Monthly Notices of the Royal Astronomical Society* 457, 217.
3. Payne, M. J., Deck, K. M., Holman, M. J., Perets, H. B. (2013). Stability of Satellites in Closely Packed Planetary Systems. *The Astrophysical Journal Letters* 775, L44.
4. Payne, M. J., Ford, E. B. (2011). An Analysis of Jitter and Transit Timing Variations in the HAT-P-13 System. *The Astrophysical Journal* 729, 98.
5. Payne, M. J., Ford, E. B., Veras, D. (2010). Transit Timing Variations for Inclined and Retrograde Exoplanetary Systems. *The Astrophysical Journal Letters* 712, L86.
6. Payne, M. J., Ford, E. B., Wyatt, M. C., Booth, M. (2009). Dynamical simulations of the planetary system HD69830. *Monthly Notices of the Royal Astronomical Society* 393, 1219.
7. Payne, M. J., Wyatt, M. C., Thébault, P. (2009). Outward migration of terrestrial embryos in binary systems. *Monthly Notices of the Royal Astronomical Society* 400, 1936.
8. Payne, M. J. (2009). The formation and evolution of planetary systems. *Ph.D. Thesis*.
9. Payne, M. J., Lodato, G. (2007). The potential for Earth-mass planet formation around brown dwarfs. *Monthly Notices of the Royal Astronomical Society* 381, 1597.

Grants and observing time

2023–2026 **A statistically sound model for astrometric uncertainties**
NASA Opportunities for Research in Planetary Defense · Co-I · $445,911

2021–2026 **Minor Planet Center Augmentation**
NASA Cooperative Agreement · PI · $938,661

2021–2026 **Minor Planet Center Operations and Development**
NASA Cooperative Agreement · PI · $15,859,297

2020–2024 **A TESS Search for Distant Solar System Objects**
NASA TESS Guest Investigator grant · PI · $250,000

2019–2023 **Dynamical Processes in the Outer Solar System**
NASA Solar System Workings grant · Co-I · $517,584

2017–2021 **Developing the Pan-STARRS Outer Solar System Pipeline for NEO Detection**
NASA Solar System Observations grant · PI · $270,999

2016–2021 **Magellan-IMACS Occultation Survey**
NASA Solar System Observations grant · PI · $318,528

2017–2018 **Observational Constraints on Planet Nine**
Smithsonian Scholarly Studies · PI · $59,100

2017–2018 **PANIC Survey: The Pan-STARRS Isotropic Centaur Survey**
Smithsonian Scholarly Studies · Co-I · $59,100

2016–2017 **The IMACS Occultation Survey for Sub-km KBOs**
Smithsonian CGPS/Pell Grant · Joint PI · $27,766

2015–2016 **A Novel Technique for Discovering Near Earth Objects**
Smithsonian IR&D Grant · Co-I · $47,363

2014–2016 **Magellan-IMACS Occultation Survey (2014B, 2015A & B, 2016A & B)**
Magellan-IMACS observing time · Co-I

2013–2017 **Novel Model for the Origin of Irregular Moons**
Bi-National Science Foundation Grant · Joint PI · $250,000

2013–2017 **Stability of Planetary Satellites in White Dwarf Systems**
Royal Society International Exchange Grant · Joint PI · $10,000

2013–2015 **HARPS-N observations (2013B, 2014A & B, 2015A)**
HARPS-N observing time · PI

2013–2014 **Keck-HIRES observations (2013A & B)**
Keck-HIRES observing time · PI · $27,000

2013–2014 **Discovering and Characterizing Circumbinary Planets and Their Satellites**
Smithsonian CGPS/Pell Grant · Co-I · $60,594

Recent talks

- 24 September 2026: Minor Planet Center report. International Asteroid Warning Network (IAWN) meeting.
- September 2026: Three guest lectures: “The Minor Planet Center”, “Planetary defense: the threat from near-Earth objects” and “Earth’s other moons: how many minimoons does Earth have right now?”. First-year seminar taught by Charles Alcock.
- 31 August 2026: An overview of the Minor Planet Center: from postwar orbit clearinghouse to modern planetary-defense infrastructure. CfA science talk, Center for Astrophysics | Harvard & Smithsonian.
- 17 August 2026: An overview of the Minor Planet Center. Catalina Sky Survey Monsoon workshop.
- 6 July 2026: An overview of the Minor Planet Center: from postwar orbit clearinghouse to modern planetary-defense infrastructure. Asteroids, Comets, Meteors (ACM) 2026.
- 7 January 2026: Solar System science with TESS. American Astronomical Society meeting (AAS 247).
- 12 December 2025: The Minor Planet Center and 3I/ATLAS. IAU Office for Astronomy Outreach: 3I/ATLAS virtual Q&A.
- 8 October 2025: Minor Planet Center activities. Space Mission Planning Advisory Group (SMPAG).
- 2 October 2025: Minor Planet Center updates. International Asteroid Warning Network (IAWN) meeting.
- 6 August 2025: Minor Planet Center processing in the Rubin era. NASA Planetary Data System Management Council.

Full list: matthewjohnpayne.github.io/talks.

Students and postdocs supervised

- **Kim Armas**, SPARK student · 2025–2026.
- **Amanda Fanale**, SPARK student · 2025–2027.
- **Milagros Tamara Giraldo**, Honors thesis student, Bridgewater State University · 2024–2026.
- **Devin Hoover**, Undergraduate researcher · 2022.
- **Jonathan Sullivan**, Undergraduate researcher · 2022.
- **Mike Alexandersen**, Postdoctoral researcher · 2020–2023.
- **Zachary Murray**, Harvard PhD student (thesis advisory committee member) · 2019–.
- **David Hernandez**, Postdoctoral researcher · 2018–2022.

- **Max Goldberg**, Undergraduate researcher · 2018.
- **Tyler Valentine**, Undergraduate researcher · 2018.
- **Sam Hadden**, Postdoctoral researcher · 2017–2018; CfA Fellow · 2018–2021.
- **Arielle Joasil**, Undergraduate, Smithsonian Latino Initiative · 2016–2017.
- **Jamie Dietrich**, Harvard senior thesis · 2016–2017.
- **Sara Khalafinejad**, Visiting PhD student, University of Hamburg · 2016.
- **Tom Leith**, Harvard senior thesis · 2015–2016.

Service

- Member of the Steering Committee of the International Asteroid Warning Network (IAWN), for the core area of data management, archiving and distribution.
- Referee for ApJ, ApJL, MNRAS, A&A and other journals.
- Reviewer on national peer-review panels for NASA (TESS, Kepler/K2, WFIRST, SSO, XRP and ATP) and the NSF.
- Member of departmental committees, including the Fellowship Selection Committee and the Scientific Computation Committee.

Computing

Python · LaTeX · Bash · C, C++ and Fortran · Git, GitHub and continuous integration · Docker